Neural Networks

MLP Neural Networks Example and Interpretation

Slides are partially adapted from notes by Vincent Cheung, Kevin Cannons and William Kaiser

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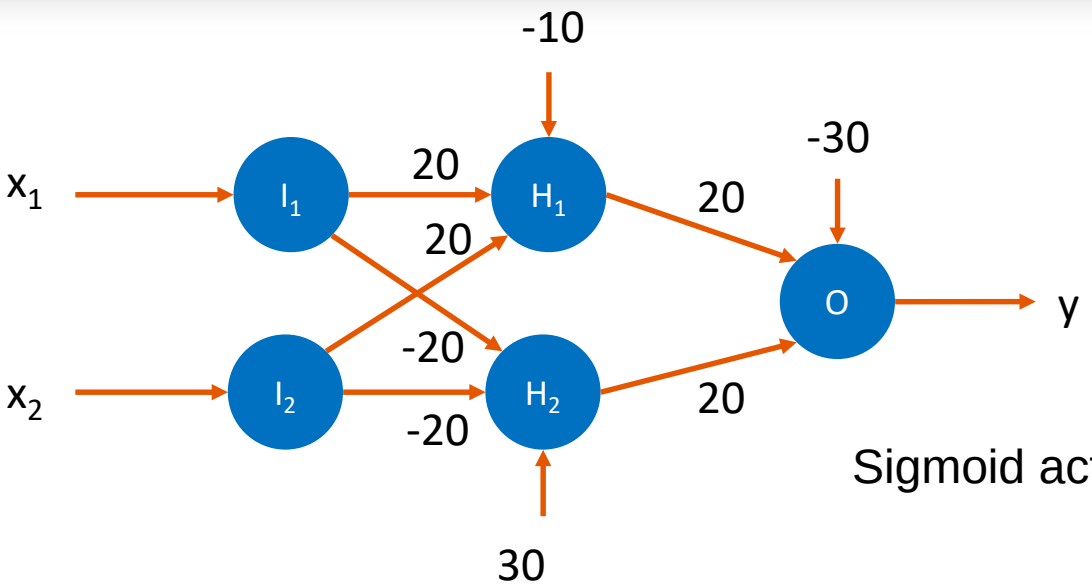
Example: Another Implementation of XOR



Sigmoid activation function for H_1, H_2 and O:

$$f(y) = \frac{1}{1 + e^{-y}}$$

X_1	X_2	H_1	H_2	O
0	0			
0	1			
1	0			
1	1			

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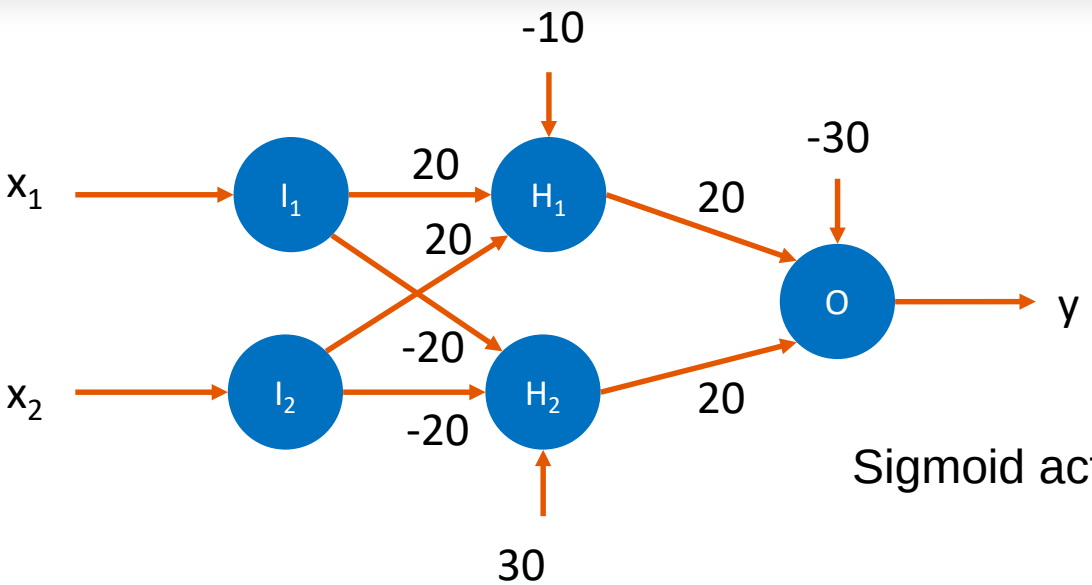

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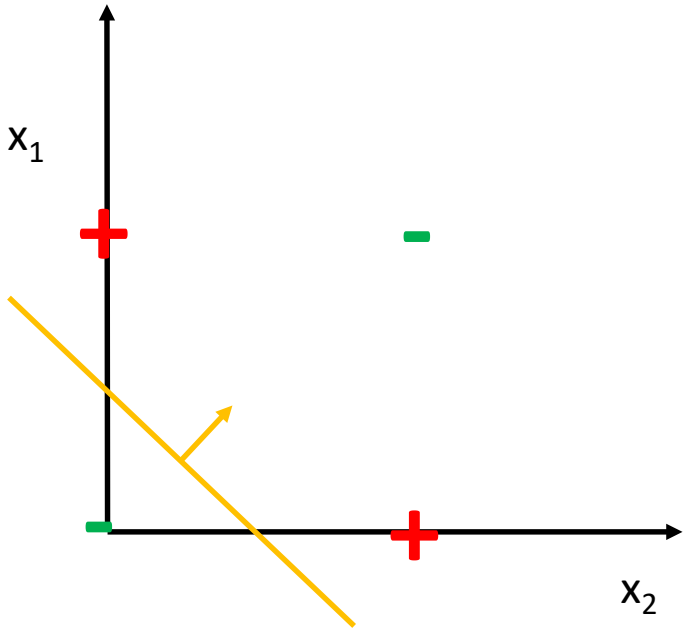
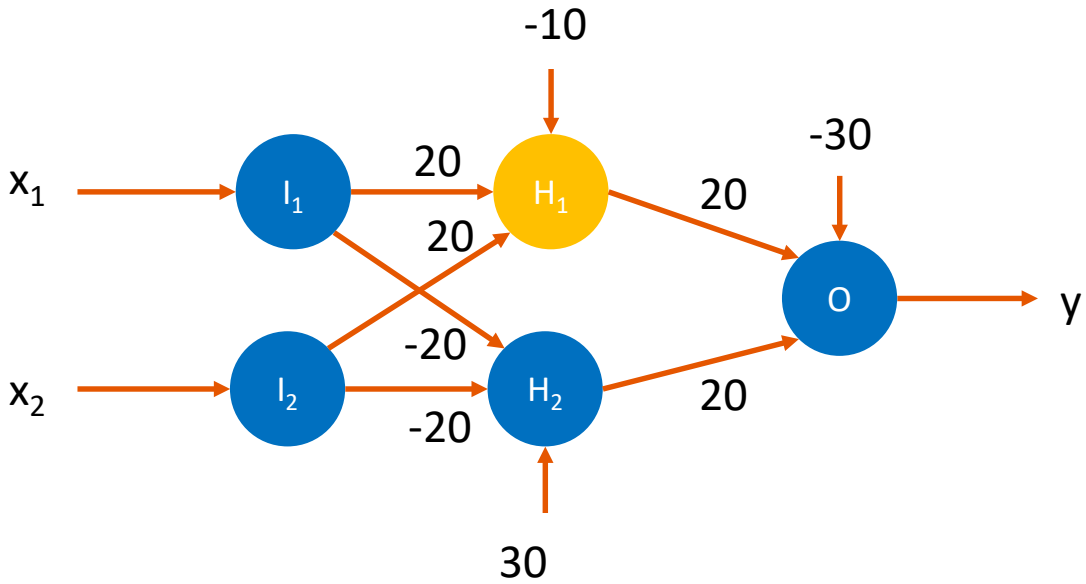
$$f(y) = \frac{1}{1 + e^{-y}}$$

X_1	X_2	H_1	H_2	O
0	0	20*0+20*0-10=-10 f(-10)=0	-20*0-20*0+30=30 f(30)=1	20*0+20*1-30=-10 f(-10)=0
0	1	20*0+20*1-10=10 f(10)=1	-20*0-20*1+30=10 f(10)=1	20*1+20*1-30=10 f(10)=1
1	0	20*1+20*0-10=10 f(10)=1	-20*1-20*0+30=10 f(10)=1	20*1+20*1-30=10 f(10)=1
1	1	20*1+20*1-10=30 f(30)=1	-20*1-20*1+30=-10 f(-10)=0	20*1+20*0-30=-10 f(-10)=0

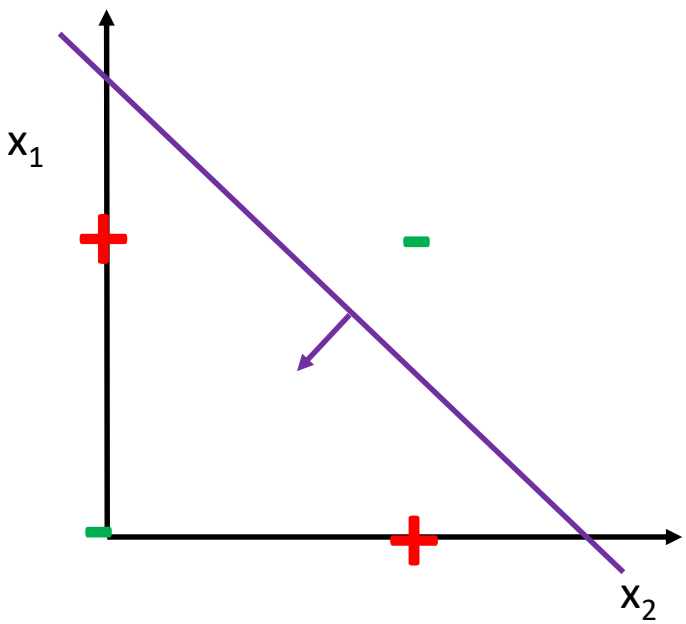
What does a single neuron do?

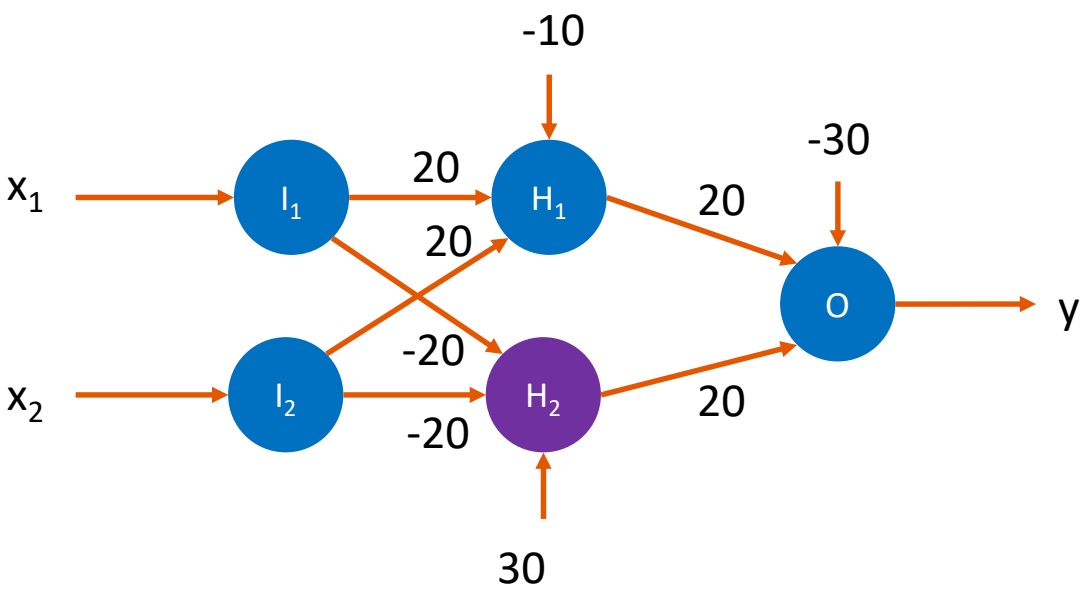




H_1: $20x_1 + 20x_2 - 10 > 0$?

What does a single neuron do? (cont'd)



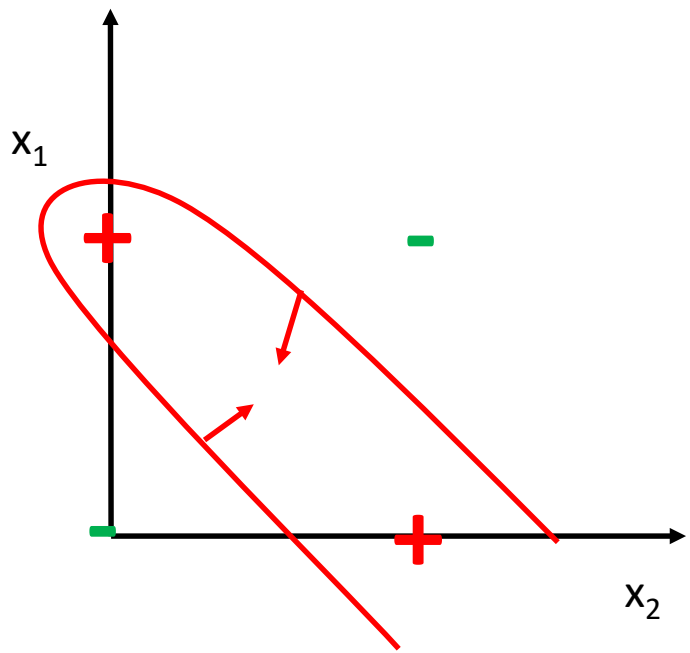


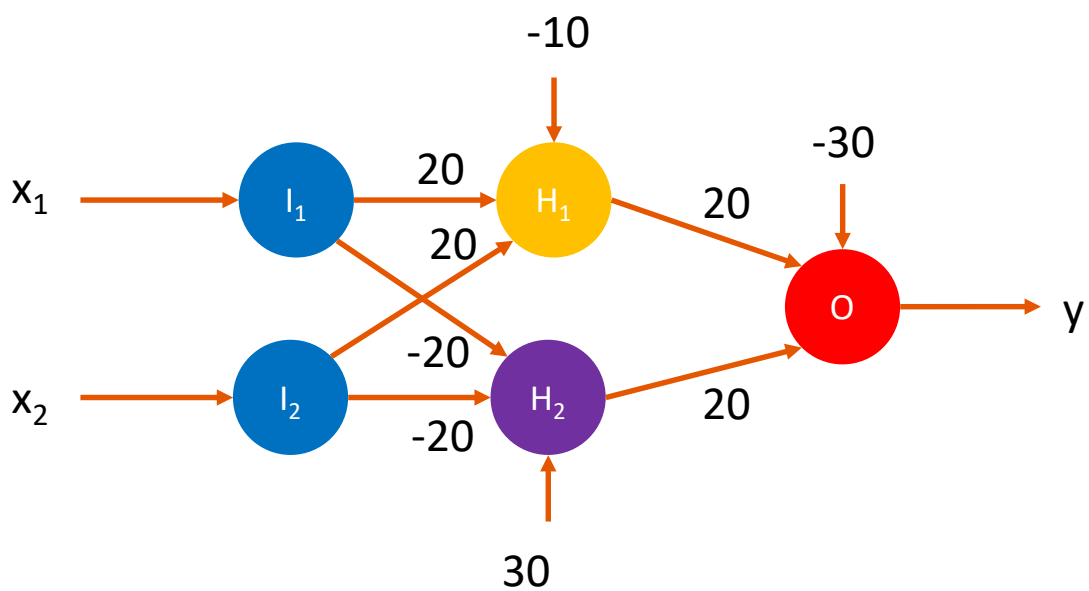
$H_2: -20x_1 - 20x_2 + 30 > 0?$

What does a single neuron do? (cont'd)

- With two inputs, a neuron can classify the data points in two-dimensional space into two kinds with a straight line.
- If you have three inputs, a neuron can classify data points in three-dimensional space into two parts with a flat plane.
- For n inputs, a neuron can divide n-dimensional space with a hyperplane.

What does multiple neurons (hidden layers) do?





H_1: $20x_1 + 20x_2 - 10 > 0$? (OR gate)

H_2: $-20x_1 - 20x_2 + 30 > 0$? (NAND gate)

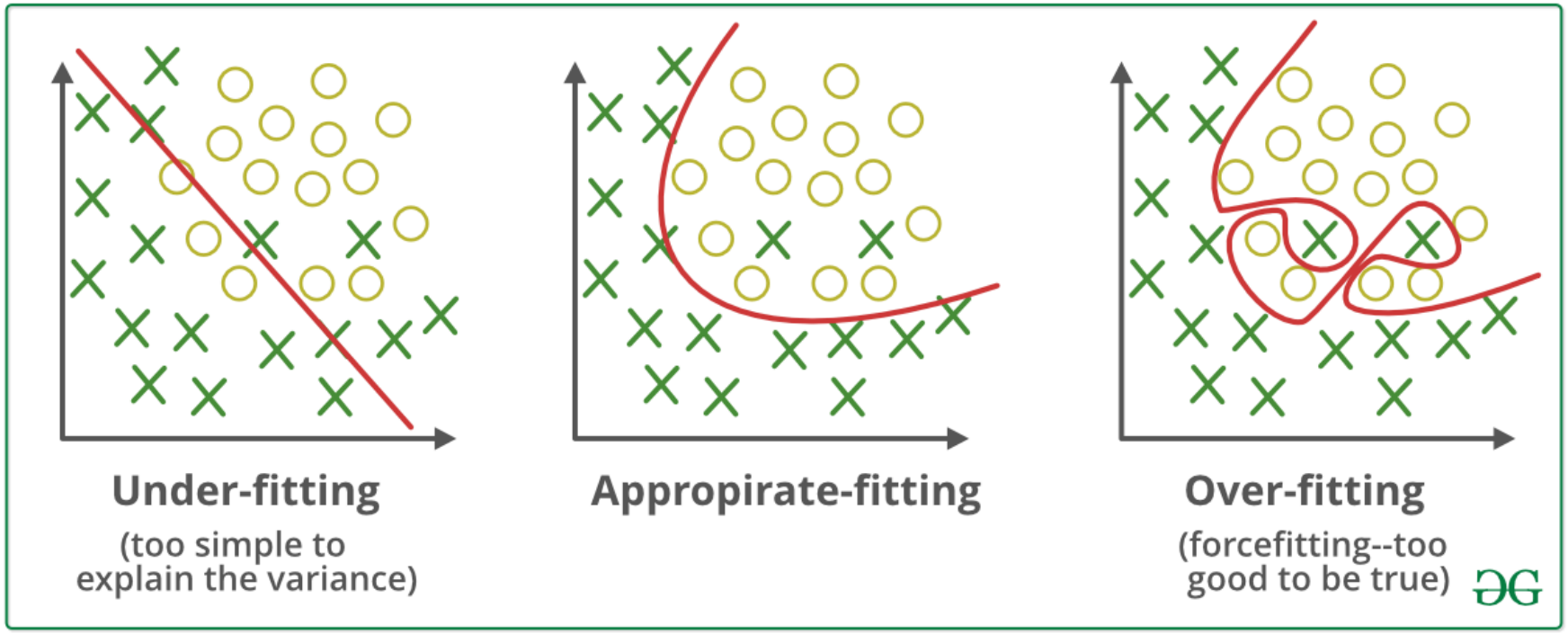
O: $20H_1 + 20H_2 - 30 > 0$? (combine H_1 and H_2 outputs together, AND gate)

What does multiple neurons (hidden layers) do? (cont'd)

- **Each neuron extract one feature from the data. Output weights from these neurons indicate the strength of their corresponding features.**
- **A composition of multiple neurons (features) can extract more sophisticated features from the dataset.**
- **Therefore, multiple neurons / hidden layer(s) can build complex (nonlinear) shapes to classify data.**
- **Try it out:**
  - http://playground.tensorflow.org/

https://developers.google.com/machine-learning/glossary

Under-fitting vs. Over-fitting



https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/

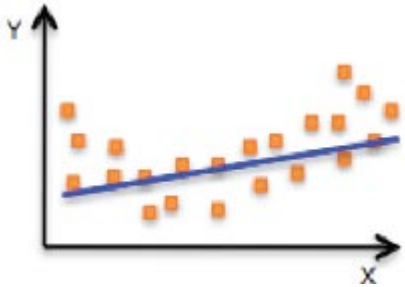
Question

The regression model in the figure is

a) Under-fitting

b) Appropriate fitting

c) Over-fitting



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